APPROVAL SHEET
承 认 书

Customer 客户名称	
Part NO. 产品型号	S8001280B1060B-32LB
Product type 产品内容	Mode: Transmissive type .Normally Black. TFT LCD Module LCD Module: Graphic 800RGB*1280Dot-matrix
Remarks 备注栏	☐APPROVAL FOR SEPCIFICATIONS ONLY ■APPROVAL FOR SEPCIFICATIONS AND SAMPLE
Signature by Customer: 客户确认签章	

Issued by	Checked by	Approved by

REVISION STATUS

Version	Revise Date	Page	Content	Modified by
V1.0	2022-7-12	-	First Issued.	

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1. General Description

* DESCRIPTION

S8001280B1060B-32LB is a color active matrix TFT (Thin Film Transistor) LCD (liquid crystal display) that uses amorphous silicon TFT as a switching device. This model is composed of a Transmissive type TFT-LCD Panel, driver circuit, back-light unit. The resolution of a 10.1" TFT-LCD contains 800 x 1280 pixels, and can display up to 16.7M colors.

*** Features**

-Low Input Voltage: IOVCC: 1.65~3.3V;VCC: 2.5~3.3V
-Display Colors of TFT LCD: 16.7M colors
- Interface: MIPI
-Internal Power Supply Circuit.

General Information Items	Specification	Unit	Note
	Main Panel		
Display area(AA)	135.36(H) x 216.576(V)	mm	-
Driver element	a-Si TFT active matrix	-	-
Display colors	16.7M	colors	-
Number of pixels	800(RGB) *1280	dots	-
Pixel arrangement	RGB vertical stripe	-	-
Pixel pitch	0.1692(H) ×0.11692(V)	mm	-
Viewing angle	FREE	o'clock	-
Drive IC	JD9365DA-H3	-	-
Display mode	Normally Black	-	-
Operating temperature	-10～+60	℃	-
Storage temperature	-20～+70	℃	-

Mechanical Information

Item		Min.	Typ.	Max.	Unit	Note
Module size	Horizontal(H)	-	143	-	mm	±0.2
	Vertical(V)	-	228.61	-	mm	±0.2
	Depth(D)	-	2.6		mm	±0.2
Weight		-	TBD	-	g	-

2. MECHANICAL SPECIFICATION

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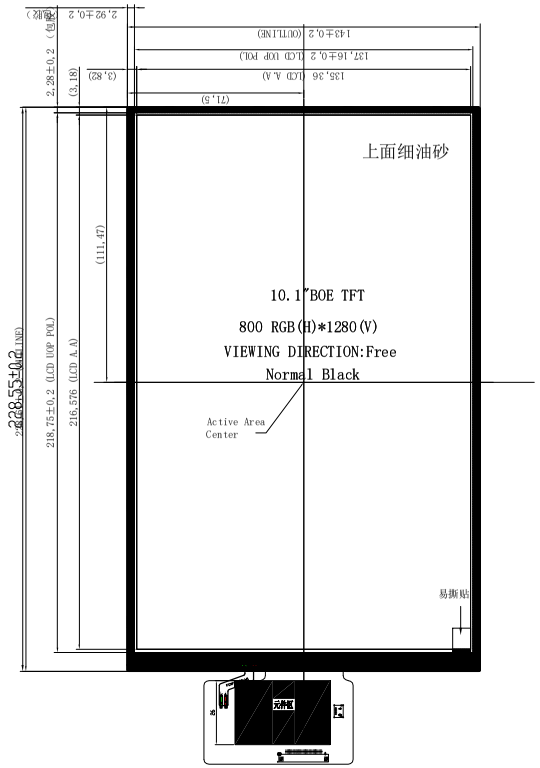
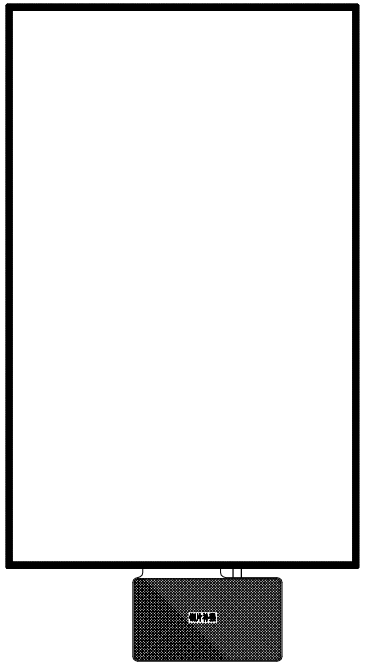
All Pages Of This Edition Approved			
Signature:		Date:	

ROHS

REV.	DESCRIPTION	REVISER	DATE
V0	NEW	JO	2024.04.26

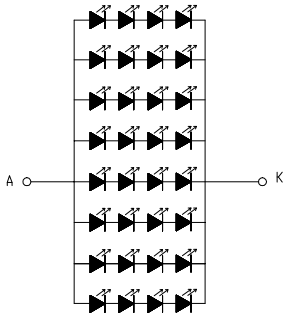




	Pin description
1	VCC 5V0
2	VCC 5V0
3	VCC 5V0
4	VCC 3V3
5	VCC 3V3
6	GND
7	GND
8	GND
9	TP SCL
10	TP SDA
11	TP INT
12	TP RESET
13	LCD BL PWM
14	NC
15	LCD RESET
16	GND
17	MIPI P2
18	MIPI N2
19	GND
20	MIPI P1
21	MIPI N1
22	GND
23	MIPI CLKP
24	MIPI CLKN
25	GND
26	MIPI P0
27	MIPI N0
28	GND
29	MIPI P3
30	MIPI N3

TP	Pin description
1	TP VDD
2	GND
3	TP RESET
4	TP SDA
5	TP SCL
6	TP_INT

LED电路图



LED CIRCUIT DIAGRAM

Vf=11.2V-12.8V(If=8*20mA)

4串8并=32LED 320nits

Specification:

1). Viewing angle: FULL VIEW
2). Display mode: TFT/Transmissive/Normal Black
3). Operating temp.: -10° C~+60° C
Storage temp.: -20° C~+70° C
4). Backlight: 32 chip White LED , In Series, If=160mA,
5). Dimensions with mark "*" are important, with mark "()" are referenced
6). According to the RoHS

Unspecified Tolerance: ±0.20	UNITS: mm	DATE:	MODEL NUMBER : S8001280B1060B-32LB	PROJECTION	SHEET:	1 Of 1
DESIGN BY:	JO	2023.10.19			DRAWING DESCRITUON:	LCM
CHECKED BY:						
APPROVED BY:						

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3.PIN DESCRIPTION

Pin NO.	Symbol	Level	Remark
1-3	VCC_5V0	H	BL POWER 5.0V
4-5	VDD	H	Logic power 3.3V
6-8	GND	L	Ground
9	TP_SCL	I/O	TP Serial Clock Line Pin
10	TP_SDA	I/O	TP Serial Data Line Pin
11	TP_INT	I/O	TP Interrupt Pin
12	TP_RESET	H/L	TP Reset Pin
13	LCD_BL_PWM	I/O	PWM control backlight
14	NC	-	Not Connect
15	LCD_RESET	H/L	LCM Reset Pin（3.3V）
16	GND	L	Ground
17	MIPI_P2	I/O	MIPI data positive signal(2+)
18	MIPI_N2	I/O	MIPI data positive signal(2-)
19	GND	L	Ground
20	MIPI_P1	I/O	MIPI data positive signal(1+)
21	MIPI_N1	I/O	MIPI data positive signal(1-)
22	GND	L	Ground
23	MIPI_CLKP	I/O	MIPI CLK positive signal(CLK+)
24	MIPI_CLKN	I/O	MIPI CLK positive signal(CLK-)
25	GND	L	Ground
26	MIPI_P0	I/O	MIPI data positive signal(0+)
27	MIPI_N0	I/O	MIPI data positive signal(0-)
28	GND	L	Ground
29	MIPI_P3	I/O	MIPI data positive signal(3+)
30	MIPI_N3	I/O	MIPI data positive signal(3-)

4. ELECTRICAL CHARACTERISTICS

4.1 ABSOLUTE MAXIMUM RATINGS

Item	Symbol	Values		Unit	Remark
		Min	Max.		
Supply Voltage for Logic circuit	VDDIO	1.7	3.0	V	
Supply Voltage for analog circuit	Vcc	3.0	3.6	V	

4.2 DC ELECTRICAL CHARACTERISTICS

4.2.1 POWER SPECIFICATION

INPUT POWER SPECIFICATIONS ARE AS FOLLOWS:

THE POWER SPECIFICATION ARE MEASURED UNDER 25℃ AND FRAME FREQUENCY UNDER 60HZ

Typical Operating Conditions (Ta=25℃)

Item	Parameter	Values			Unit	Remark
		Min	Typ	Max.		
VCI	Vcc	3.0	3.3	3.6	V	
VDD3	Iovcc	1.7	1.8	3.0	V	
TFT Gate ON Voltage	Pixel turn on voltage	14	15	16	V	
TFT Gate OFF Voltage	Pixel turn off voltage	-14	-13	-12	V	
VSP	Negative AVEE voltage	4.8	5	6	V	
VSN	Postive AVDD voltage	-6	-5	-4.8	V	

4.2.2 BACKLIGHT UNIT (GND=0V)

Item	Symbol	Values			Unit	Remark
		Min	Typ	Max.		
Forward supply Voltage	V_f	11.2	12.0	13.2	V	
Forward supply Current	I_f	-	160	-	mA	
LCM Luminance	Lv	280	320	-	cd/m2	Ib=160mA
Uniformity	/	75	80		%	-

5. OPTICAL CHARACTERISTICS

The optical characteristics are measured under stable conditions at 25℃ (Room Temperature) :

Item		Symbol	Conditions	Min.	Typ.	Max.	Unit	Note
Viewing Angle		θ_R	Horizontal (Right) CR = 10	80	85	-	degree	1, 4
		θ_L	(Left)	80	85	-		
		ϕ_H	Vertical (Upper) CR = 10	80	85	-		
		ϕ_L	(Lower)	80	85	-		
Transmittance		%		4.63%	5.45%	NA		C-light
Response Time		T_{RT}	Rising + Falling	-	30	35	msec	1, 3
Color / Chromaticity Coodinates	Red	Rx	CIE 1931		0.630		-	C-light
		Ry			0.335			
	Green	Gx			0.288			
		Gy			0.584			
	Blue	Bx			0.132			
		By			0.175			
	White	Wx			0.316			
		Wy			0.369			
NTSC		%		-	56.5	-		

*Note (1) Definition of Contrast Ratio (CR):

The contrast ratio can be calculated by the following expression.

Contrast Ratio (CR) = L63 / L0

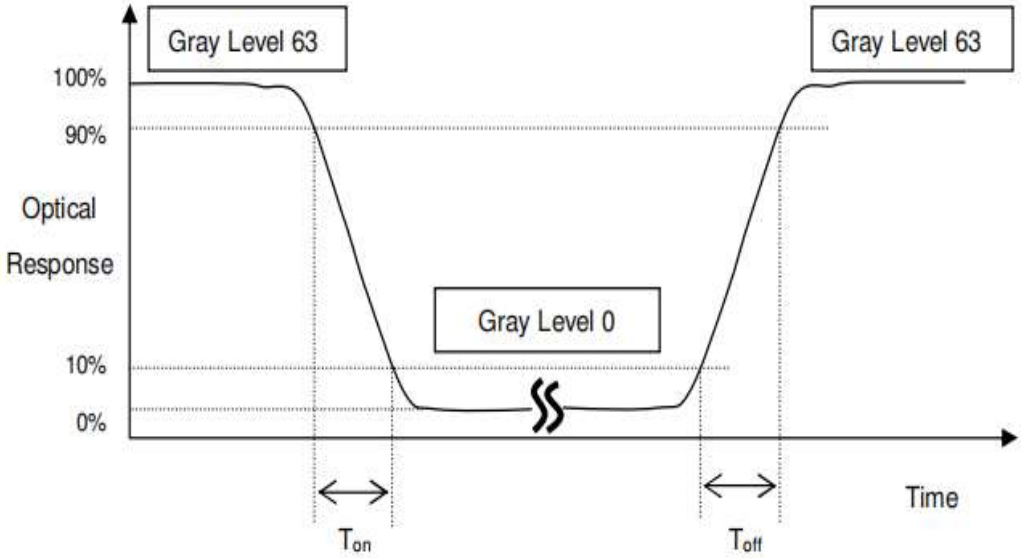
L63: Luminance of gray level 63

L 0: Luminance of gray level 0

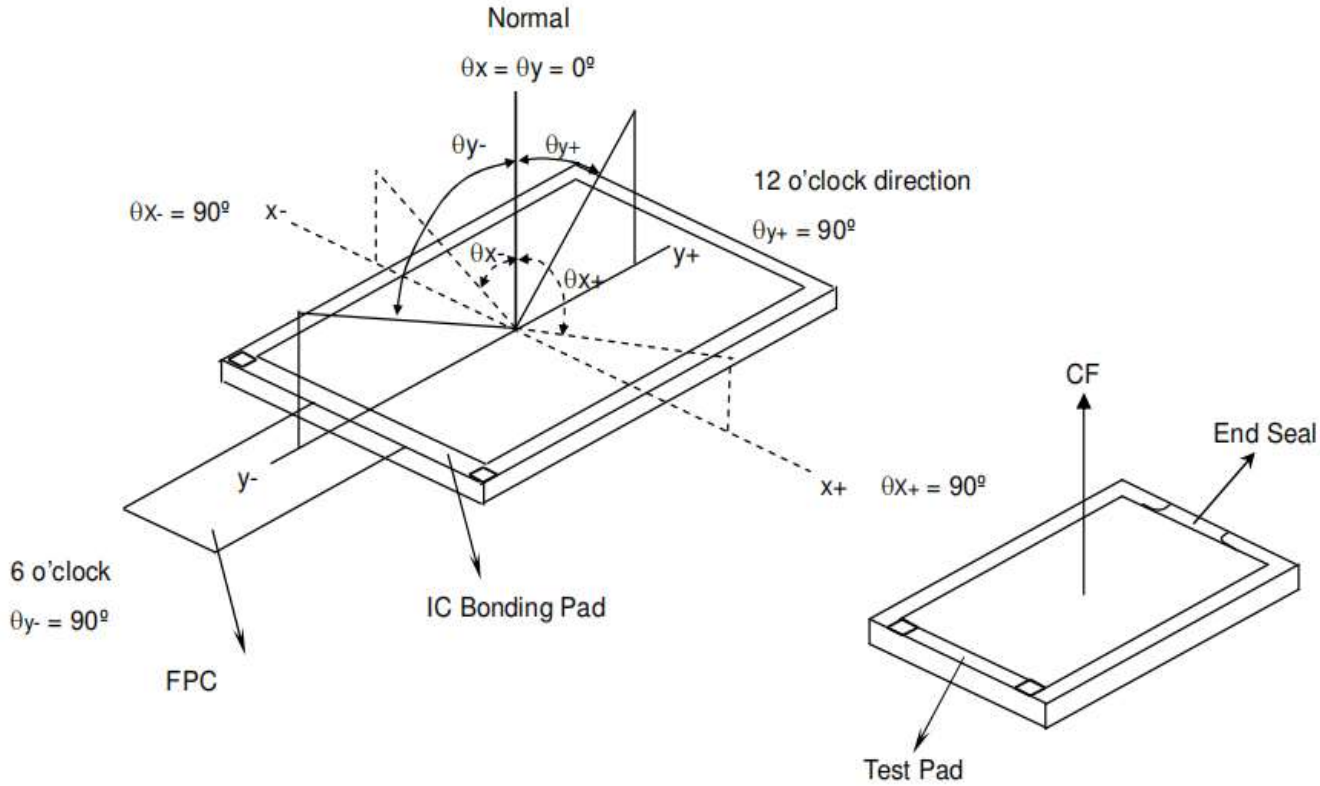
CR = CR (5)

CR (X) is corresponding to the Contrast Ratio of the point X at Figure in Note (5).

*Note (2) Definition of Response Time (T_{on}, T_{off}):

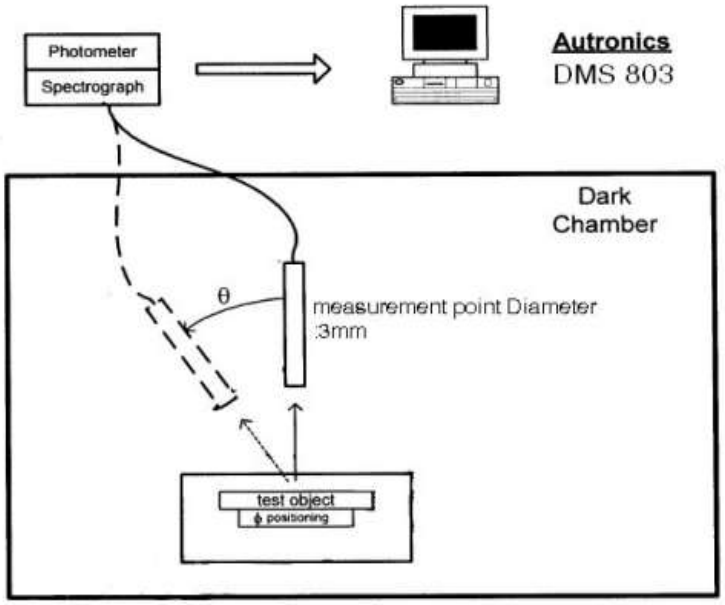


*Note (3) Definition of Viewing Angle

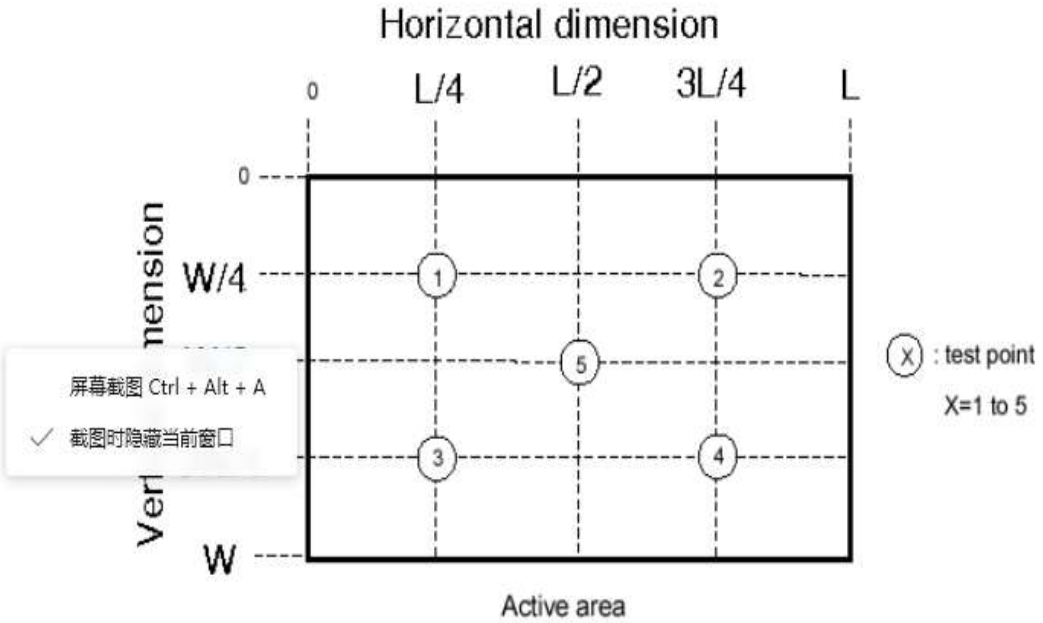


*Note (4) Measurement Set-Up:

The LCD module should be stabilized at a given temperature for 20 minutes to avoid abrupt temperature change during measuring. In order to stabilize the luminance, the measurement should be executed after lighting Backlight for 20 minutes in a windless room.



*Note (5)



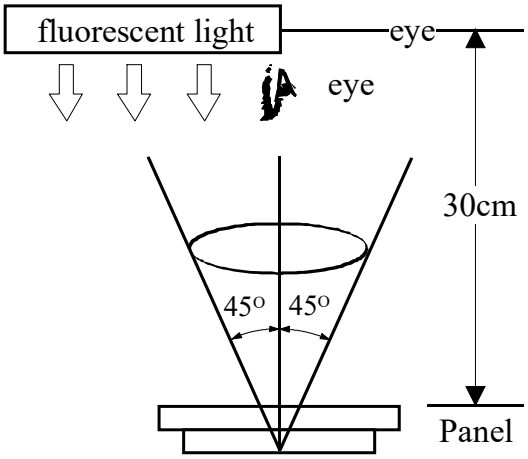
屏幕截图 Ctrl + Alt + A
截图时隐藏当前窗口

6. QUALITY SPECIFICATIONS

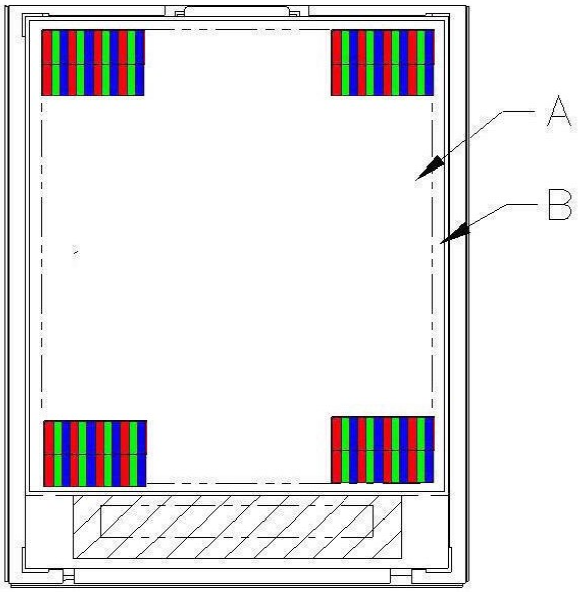
6.1 INSPECTION CONDITION

(1) Inspect under 300~500Lux fluorescent light, leaving 30~35cm between panels and eyes, and between panels and lights.

(2) Inspection condition is 23±5°C, 50±20%RH maximum.



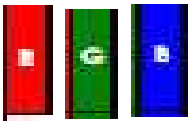
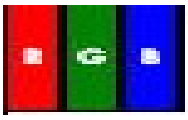
6.2 DEFINITION OF AREA



A Area : Viewing area.

B Area : Out of viewing.(outside viewing area)

6.3 INSPECTION SPECIFICATION

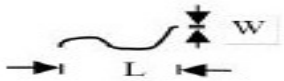
NO	Item	Acceptable specification	Judgment Criterion
1	Electrical Testing	**1-1 sub pixel classification** ● Sub Pixel: Number of sub pixel doesn't exceed one dot.  Sub Pixel (Dot) a> Dark dot ----one Allowed b> Bright dot ---- one Allowed	N≦1
		● Pixel : Three dots link together doesn't exceed ones  Pixel	N≦0
		1-2 Leakage to light ● Leakage to light be not allowed.	N=0
		1-3 Picture to shake ● Picture had shake, twinkle and noise etc. instable of defect that be not allowed.	N=0
		1-4 Function ● No display or No function. ● Source Line, Gate Line. ● Contrast Ratio ● Current consumption exceeds product specifications. ● Display malfunction.	N=0
2	Mechanical Dimension	2-1 Mechanical Dimension exceeds product specifications. 2-2 Out of frame and boss of plastic changed shape that be not allowed.	N=0

NO	Item	Acceptable specification	Judgment Criterion
3	Cosmetic Inspection		

3-1 Blemish: Line shapes of defect

Length	Width	Acceptable number	Mini. space
---	W≦0.03	Ignore	5 m m
L≦2.5	0.03<W≦0.05	3	
L≦2.5	0.05<W≦0.1	2	
--	W＞0.1	Not allowed	---

L: length(mm)
W: width(mm)



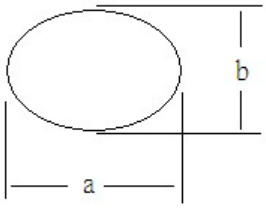
3-2 Blemish: dot shapes of defect.

Dimension	Acceptable number	Mini. Space
Φ≦0.10	Ignore	---
0.10＜Φ≦0.25	2	10 m m
0.25＜Φ≦0. 3	1	
Φ＞0.3	0	---

3-3 Polarizer Bubble

Dimension	Acceptable number	Mini. Space
Φ≦0.20	Ignore	---
0.20＜Φ≦0.30	2	15 m m
Φ＞0.30	0	---

Foreign Substances



Φ=(a+b)/2

<table>
<tr><th>NO</th><th>Item</th><th colspan="4">Acceptable specification</th><th>Judgment Criterion</th></tr>
<tr><td rowspan="8">3</td><td rowspan="8">Cosmetic Inspection</td><td colspan="4">3-4 Scratch
● Sensate scratch not allowed.
● Impassive scratch as below.
Unit:mm</td><td rowspan="8"></td></tr>
<tr><td>Length</td><td>Width</td><td>Acceptable number</td><td>Mini. space</td></tr>
<tr><td>-----</td><td>W≦0.03</td><td>Ignore</td><td rowspan="3">5 m m</td></tr>
<tr><td>L≦2.5</td><td>0.03<W≦0.05</td><td>3</td></tr>
<tr><td>L≦2.5</td><td>0.05<W≦0.1</td><td>2</td></tr>
<tr><td>----</td><td>0.1＜W</td><td>Not allowed</td><td rowspan="2">---</td></tr>
<tr><td>L>2.5</td><td>----</td><td>Not allowed</td></tr>
<tr><td colspan="4"></td></tr>
<tr><td>4</td><td>Package</td><td colspan="4">4-1 Mixed product types
4-2 Shipping q’ty should be the same as “shipping notice form” q’ty.
4-3 Outer box can’t broken.</td><td>N=0</td></tr>
</table>

7. RELIABILITY

Test Item	Test Condition
High Temperature Operation	60℃ for 24 hours
Low Temperature Operation	-10℃ for 24 hours
High Temperature Storage	70℃ for 24 hours
Low Temperature Storage	-20℃ for 24 hours
High Temperature Operation Humidity Operation	50℃, 90%RH for 48 hours
Thermal Shock	-10°storage one hour,rise to 60° within 15s,high temperature one hour, drop to 30° within 15s, circulate 10 repeatedly
Vibration Test (No Operation)	Frequency: 10~55Hz Amplitude:1.0mm Sweep Time: 11min Test Period: 6 Cycles for each direction of X, Y, Z
Static electricity test	Touch 6KV,air touch 8KV

8. HANDLING PRECAUTION

8.1 SAFETY

(1) Do not swallow any liquid crystal, even if there is no proof that liquid stal is poisonous.
(2) If the LCD panel breaks, be careful not to get liquid crystal to touch your skin.
(3) If skin is exposed to liquid crystal, wash the area thoroughly with alcohol or soap.

8.2 STORAGE CONDITIONS

(1) Store the panel or module in a dark place where the temperature is 23±5℃ and thehumidity is below 50±20%RH.
(2) Store in anti-static electricity container.
(3) Store in clean environment, free from dust, active gas, and solvent.
(4) Do not place the module near organics solvents or corrosive gases.
(5) Do not crush, shake, or jolt the module.

8.3 HANDLING PRECAUTIONS

(1) Avoid static electricity which can damage the CMOS LSI.
(2) The polarizing plate of the display is very fragile. So, please handle it very carefully.
(3) Do not give external shock.
(4) Do not apply excessive force on the surface.
(5) Do not wipe the polarizing plate with a dry cloth, as it may easily scratch the surface of plate.
(6) Do not use ketonics solvent & Aromatic solvent, use with a soft cloth soaked with acleaning naphtha solvent.
(7) Do not operate it above the absolute maximum rating.
(8) Do not remove the panel or frame from the module.

8.4 WARRANTY

The period is within twelve months since the date of shipping out under normal using and storage conditions.